

Anirudh Rayas

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Postdoctoral researcher in machine learning and AI with experience in healthcare and clinical AI, agentic AI-driven clinical reasoning systems, and multimodal LLM evaluation and calibration. Ph.D. in graph machine learning with publications at leading venues including NeurIPS, ICML, ICASSP, ISIT, WACV, IEEE Transactions, and Frontiers. Current research spans graph representation learning and graph transformer methods for clinical prediction, multi-agent orchestration for clinical decision support and biomedical reasoning, and calibration of MLLM-as-a-Judge systems for reliable evaluation in high-stakes healthcare applications. Seeking to leverage AI and machine learning to accelerate drug discovery and advance data-driven therapeutic development.

Professional Experience

Postdoctoral Research Scholar — Machine Learning & AI

Sep 2025 – Present

Arizona State University

- Developed a heterogeneous temporal hypergraph transformer (HT-HGNN) for longitudinal patient modeling on **MIMIC-III/IV** electronic health records (~360K patients), integrating diagnosis (ICD), procedure, medication, and ATC drug ontology data into a graph representation-learning framework to jointly predict mortality, readmission, length of stay, and drug recommendations.
- Achieved **state-of-the-art** performance across four clinical prediction tasks, improving mortality prediction by 30%, readmission prediction by 24.7%, and overall multi-task performance by 10.2% relative to leading baselines (*manuscript under preparation*).
- Implemented distributed computing strategies to scale HT-HGNN training across multi-GPU clusters, accelerating experimentation and reducing training time by 40%.
- Designed a LangGraph-based multi-agent clinical reasoning system that converts model outputs into clinician-facing reports, explains risk predictions, evaluates medication recommendations, and flags uncertain or miscalibrated cases for review.
- Developed calibration and conformal-prediction methods for multimodal LLM outputs (“MLLM-as-a-judge”), producing distribution-free uncertainty estimates with applications in safety-critical healthcare AI.
- Lead and mentor graduate students across topics in clinical AI, graph representation learning, multimodal LLM evaluation, and uncertainty quantification for healthcare and scientific applications.

AI/ML R&D Consultant

Jul 2025 – Present

Myna Voice Labs

- Built end-to-end ML pipelines for early Parkinson’s disease detection from real-world voice recordings, developing foundation-model-based classifiers that achieved >90% diagnostic accuracy.
- Developed uncertainty quantification methods based on Bayesian credible intervals, enabling calibrated confidence estimates and risk-aware prediction reporting for clinician-facing decision support.
- Collaborated with engineering and product teams to transition predictive models following HIPAA compliance, toward production systems, improving reproducibility, inference reliability, and deployment readiness for clinical applications.

Research Intern — Applied Mathematics & Machine Learning

May – Aug 2024

Los Alamos National Laboratory

- Developed sample-efficient structure-learning algorithms for stochastic networked systems under partial and noisy observations, with applications to physical and biological networks.
- Established recovery guarantees in high-dimensional regimes; results published at ISIT 2025 and IEEE TSIPN 2025.

Featured Project

HT-HGNN + LangGraph Agentic Clinical Decision Support

[anirudhrayas/hthgnn-langchain](#) 2025

- Developed a heterogeneous temporal hypergraph transformer (HT-HGNN) trained on ~59K hospital admissions (MIMIC-III) for mortality prediction, 30-day readmission prediction, length-of-stay estimation, and drug recommendation (average AUC = 0.822).
- Built a retrieval-augmented clinical reasoning pipeline that retrieves clinically similar patients and constructs structured patient context from diagnoses, medications, outcomes, and longitudinal EHR records.

- Designed a LangGraph-based multi-agent workflow comprising specialized mortality, readmission, length-of-stay, and medication recommendation agents that translate model predictions into clinician-facing risk assessments and treatment recommendations.
- Implemented synthesis and critique agents that perform consistency checks, identify contraindications and unsafe medication combinations, flag calibration conflicts, and generate clinician-facing reports with uncertainty-aware recommendations.

Education

Ph.D., Electrical, Computer and Energy Engineering Arizona State University (M.S. awarded en route, 2024)	2020 – 2025
B.E., Electronics and Communication Engineering PES University	2014 – 2018

Skills

Languages	Python (Advanced), SQL, Bash
ML Frameworks	PyTorch, TensorFlow, DGL, PyTorch Geometric, GPyTorch, HuggingFace Transformers, scikit-learn, NumPy, SciPy, Pandas
Deep Learning	Graph Representation Learning (Graph Neural Networks, Graph Transformers, Hypergraph Neural Networks, Equivariant GNNs), Knowledge Graphs, Graph Structure Learning, Probabilistic Graphical Models, Multi-task Learning, Multimodal Large Language Models
AI & Agents	LangChain, LangGraph, Retrieval-Augmented Generation (RAG), MLLM-as-a-judge evaluation, Agentic AI orchestration
Statistical Inference	Uncertainty Quantification, Conformal Prediction, Bayesian Inference, High-Dimensional Statistical Learning, Biostatistics and Bioinformatics
Healthcare AI	Electronic Medical Records, Precision Healthcare, Clinical Decision Support, ICD-9 Medical Codes, ATC Drug Ontologies
Engineering	Cloud & Infrastructure (Docker, AWS, GCP, Linux, HPC), MLOps (Weights & Biases, MLflow, Git, Conda, CI/CD Pipelines), Data Visualization, Distributed (multi-GPU) Training, CUDA

Selected Publications

- **On the Calibration of Multimodal LLMs as Judges**
Submitted to EMNLP 2026
- **PCL: Partitioned Continual Learning via Unsupervised Latent Experts for Audio Classification**
ICML 2026
- **Improving Shape Bias in Learnable Geometric Representations**
WACV 2026
- **Structure Learning in Gaussian Graphical Models from Glauber Dynamics**
ISIT 2025
- **Learning Networks from Wide-Sense Stationary Stochastic Processes**
IEEE Transactions on Signal and Information Processing over Networks, 2025
- **Orthogonality and graph divergence losses promote disentanglement in generative models**
Frontiers in Computer Science 2024
- **Differential Analysis for Networks Obeying Conservation Laws**
ICASSP 2023
- **Learning the Structure of Large Networked Systems Obeying Conservation Laws**
NeurIPS 2022

Awards & Honors

NeurIPS Scholar Award	2022
Google CS Research Mentorship Program	2022
Best Presentation Award, Information Theory and Applications (ITA) Workshop	2025